4.5 Comparable measures and comparison criteria

Every method is mapped to the same objects: a VAR on which the same GFEVD connectedness is computed; a directed Granger network (sum over lags of the absolute coefficients); and contemporaneous and long-run partial-correlation networks (for DY and the lasso VAR, implied by the estimated VAR). We compare methods on three criteria:

1. *Density*: share of non-zero links in each network; for the dense GFEVD table, share of pairwise shares above $1/N$; and total connectedness.

2. *Agreement*: correlation of total and net connectedness series across methods; per-window Spearman correlation between the off-diagonal entries of two methods' matrices; Jaccard similarity of their strongest 10% of links.

3. *Stability*: variability of total connectedness, and Spearman correlation and top-link Jaccard similarity between consecutive windows' matrices.

Finally, the share of DY connectedness that disappears once the common factor is removed is $1 - C^{\text{FNETS}}/C^{\text{DY}}$, averaged over windows.

4.6 Robustness

We rerun all methods with windows of 78 and 130 weeks, horizons of 2 and 8 weeks, without SPY, and excluding 2020. For the last, every rolling window containing any week of 2020 is dropped (the resulting gap is not interpolated), and static full-period connectedness is estimated separately for 2011–2019 and 2021–2026.

5 Results

5.1 Full-sample networks

Table 4 (appendix) reports static connectedness estimated once on the whole sample; these figures differ from the averages over rolling windows reported in Table 2 and below (e.g. 81.9% against 81.0% for Diebold–Yilmaz, 34.2% against 47.2% for FNETS), because a single long-sample estimate is not the mean of short-window estimates. The two dense estimators give almost the same total connectedness (Diebold–Yilmaz 81.9%, lasso VAR 82.1%), NETS gives 73.5%, and the idiosyncratic component identified by FNETS gives only 34.2%. The heatmap of Figure 3(a) shows that Diebold–Yilmaz connectedness is spread widely across all pairs, with only mild regional clustering: the strongest pairs are Korea–Taiwan and the Indonesia–Malaysia–Philippines–Thailand block, and South Africa is a common source of shocks for almost every market. Once the common factor is removed (Figure 3(b)), the remaining pairwise shares are concentrated in a few pairs (Korea–Taiwan above all), and the full-sample idiosyncratic partial-correlation network is empty (Figure 4, right panel). This can mean either that, once the common factor is removed, there is little network structure left among these markets, or that the thresholded estimator is too strict to detect it with the available data; in rolling 104-week windows the idiosyncratic network is sparse (7% of possible long-run links on average) but not empty. We cannot distinguish the two explanations and list this among the limitations, together with the positive-definite adjustment described in Section 4.4. NETS retains 92 long-run partial-correlation links (60% of all possible pairs). The lagged (Granger) dependence is very weak in all sparse estimators: lasso, NETS and FNETS keep only 2.5%, 1.7% and 0.3% of the possible directed lag links in the rolling windows, so connectedness at the weekly frequency is overwhelmingly contemporaneous.

5.2 Connectedness over time

Figure 1 shows the rolling total connectedness of each method. Diebold–Yilmaz connectedness moves between about 72% and 89%; the lasso VAR tracks it almost exactly (correlation 0.98). Because each estimate is dated at the end of its 104-week window, an episode affects the series for two years after it occurs. Connectedness is high at the start of the sample (88%), falls to about 76% in 2014–2015, rises to about 87% in 2016–2017, is around 80% in 2018–2019, and reaches its maximum (about 89%) for windows that contain the March 2020 crash, falling back to the mid-70s after those weeks leave the window (2022–2023). NETS follows the same low-frequency pattern at a lower level, mostly between 46% and 78% (5th and 95th percentiles), but is much noisier from one window to the next and drops to about 32% in a few windows. FNETS idiosyncratic connectedness is flatter, mostly between 36% and 53%, with occasional one-window drops to about 31% (15 of 179 windows fall below 40%; see the discussion of stability below), and it does not reproduce the 2020 peak. Figure 2 helps to read this: the share of variance that FNETS attributes to the common component, between 47% and 75%, moves together with Diebold–Yilmaz total connectedness (correlation 0.95). Part of this co-movement is mechanical, since both measures increase when the correlations among all series increase, so we read it as evidence that the two move together and not as a causal statement.

5.3 Net transmitters and receivers

Figure 5 reports average net connectedness by country. In the Diebold–Yilmaz and lasso systems the main net transmitters are South Africa (+22 percentage points), South Korea (+16) and Taiwan (+12), and the main net receivers are Turkey (−22), Vietnam (−21) and Argentina (−10). NETS gives the same ranking at about half the magnitude (South Africa +10, Korea +9, Taiwan +8; Vietnam −11, Turkey −11, Argentina −5). The ranking of the countries' average net position is almost identical across DY, lasso and NETS (Spearman correlation between 0.96 and 0.99) but differs for FNETS (0.30 with DY). In the idiosyncratic network Taiwan (+9) and Indonesia (+8) are the main transmitters, Turkey is still the main receiver (−6), and Argentina is close to neutral (+1.6). Argentina is thus a receiver of the common component of emerging-market returns rather than of idiosyncratic shocks. The receivers are the markets least correlated with the rest: the average correlation of Turkey, Vietnam and Argentina with the other markets (0.39, 0.41 and 0.47) is the lowest in the sample, against 0.61 for South Africa. These directional results call for some caution. All ETFs trade in New York, while the Asian markets (and, for part of the day, the South African and Polish markets) are already closed; a US-listed fund can therefore react to news before its local market does, and the timing can shift the estimated direction of spillovers. Weekly returns reduce this effect but do not remove it, so the transmitter role of Korea, Taiwan and South Africa may partly reflect trading hours.

5.4 Comparison of methods

Table 2 summarizes the three criteria.

Density. The Diebold–Yilmaz VAR is dense by construction. The sparse estimators remove almost all lagged links, and NETS and FNETS also sparsify the contemporaneous structure: the share of non-zero long-run partial correlations is 100% for DY and lasso (which are implied by a full innovation covariance), 52% for NETS and 7% for FNETS. Among the generalized variance shares themselves, 35% (DY) and 34% (lasso) exceed the equal-share benchmark $1/N$, against 16% for both NETS and FNETS.

Agreement. Total connectedness is highly correlated between DY and the lasso VAR (0.98) and between DY and NETS (0.89). The per-window Spearman correlation of the variance-decomposition entries with DY is 0.93 for lasso and 0.80 for NETS, and the overlap of the

Table 2. Comparison criteria across methods: averages over 179 rolling windows ($w = 104$, $H = 4$), not full-sample estimates (those are in Table 4). Densities in %; agreement measured against DY; stability is the mean Spearman correlation (variance decomposition) and top-10% Jaccard similarity (long-run partial correlations) between consecutive windows.

Method	Total conn. (%)	Granger dens. (%)	LR p.corr. dens. (%)	Corr. total w/ DY	Spearman GFEVD w/ DY	Stab. GFEVD	Stab. LRPC top
DY	81.0	100.0	100.0	–	–	0.97	0.61
Lasso VAR	80.0	2.5	100.0	0.98	0.93	0.97	0.64
NETS	61.5	1.7	51.9	0.89	0.80	0.93	0.72
FNETS (idio.)	47.2	0.3	7.3	0.00	-0.06	0.82	0.12

strongest 10% of links (Jaccard) is 0.61 and 0.51. The comparison with FNETS is of a different kind: FNETS measures connectedness of the idiosyncratic component only, after the common factor that drives most of the co-movement has been removed, whereas the other three methods measure connectedness of the returns themselves. A correlation close to zero between the FNETS and DY series (0.00; 0.04 and 0.08 with lasso and NETS) and a low overlap of strongest links (Jaccard 0.16) are therefore expected by construction and should not be read as disagreement about the same object. Agreement is lower for the networks that are sparse by design: for long-run partial correlations the Spearman correlation with DY is 0.33 (lasso) and 0.36 (NETS), while lasso and NETS agree more with each other (0.66). The Granger networks barely agree (Spearman at most 0.23), as expected given how few lag links survive.

Stability. The standard deviation of total connectedness across windows is 5.1 (DY), 5.7 (lasso), 11.0 (NETS) and 4.6 (FNETS), but its mean absolute change from one window to the next is 0.7, 0.8, 4.6 and 3.0: the dense estimators move smoothly, while NETS and FNETS jump. The correlation between consecutive windows' variance-decomposition matrices is 0.97 (DY), 0.97 (lasso), 0.93 (NETS) and 0.82 (FNETS). For long-run partial-correlation links the overlap of the strongest 10% between consecutive windows is 0.61, 0.64, 0.72 and 0.12, so NETS has the most persistent sparse structure and the FNETS idiosyncratic network is the least stable. The sharp one-window drops of the FNETS series coincide with windows in which the penalized idiosyncratic VAR retains several coefficients (correlation of -0.72 between total connectedness and the number of non-zero coefficients); we therefore read them as estimation noise in short windows and not as events.

5.5 Common factor versus idiosyncratic connectedness

FNETS selects one common factor in every window, which explains on average 58% of the variance of the series. Total connectedness of the idiosyncratic VAR is 41% lower than that of the DY VAR on average (median 42%; between 26% and 61% across windows). The loss is stable across specifications (Table 3): 38% with 78-week windows, 43% with 130-week windows, 42% without SPY and 41% when windows that touch 2020 are removed, and it does not depend on the horizon. In the subsamples (Table 4), DY connectedness is 83.0% in 2011–2019 and 73.5% in 2021–2026, NETS falls from 69.6% to 56.0%, while idiosyncratic FNETS connectedness rises slightly (34.1% to 39.6%) as the common-factor variance share falls from 58% to 47%. A large part of the variation of DY connectedness over time is therefore variation in the weight of the common factor.

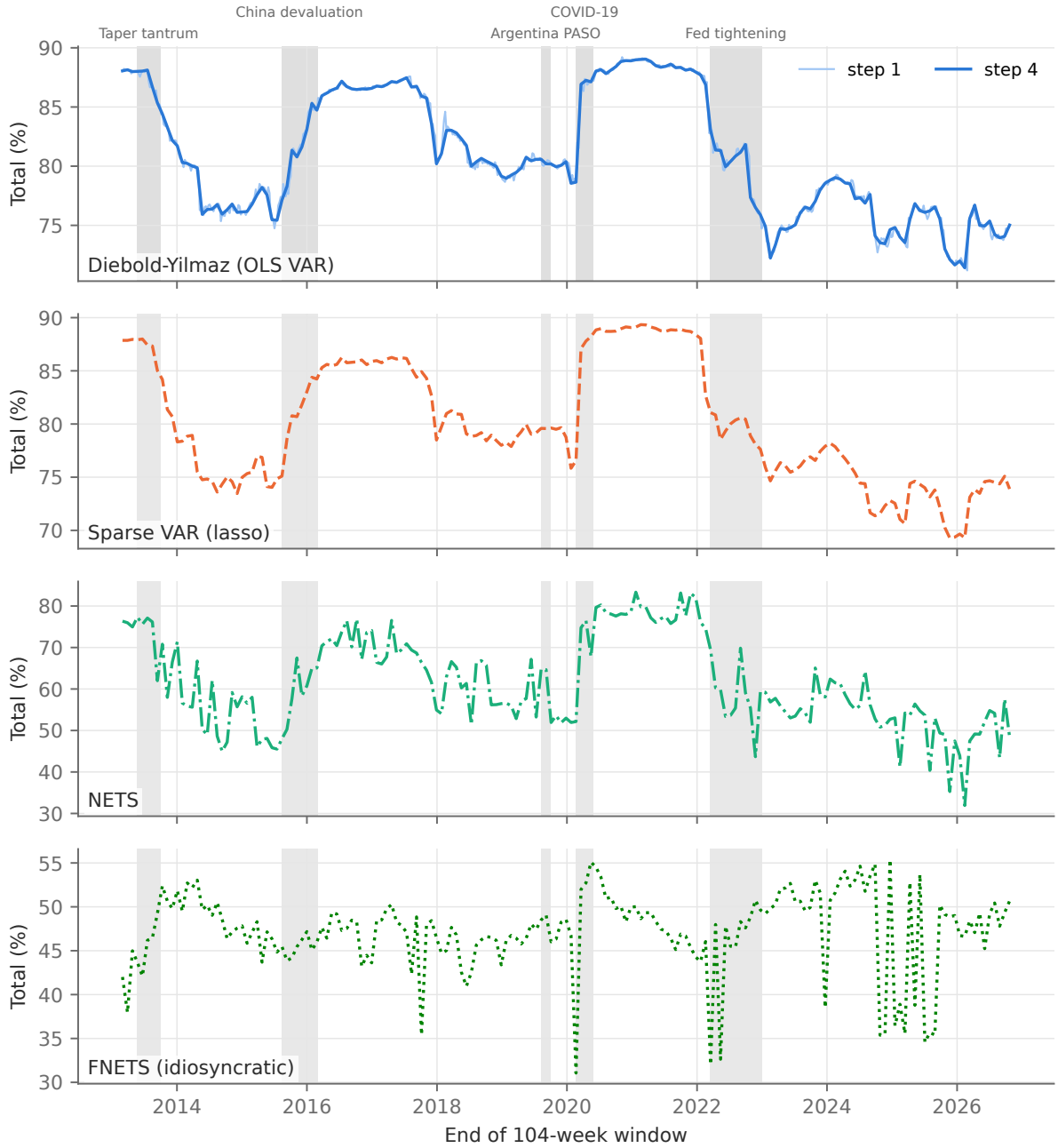


Fig. 1. Total connectedness over time by method (104-week rolling windows, $H = 4$, step 4). Estimates are dated at the end of the window. Shaded areas mark events.

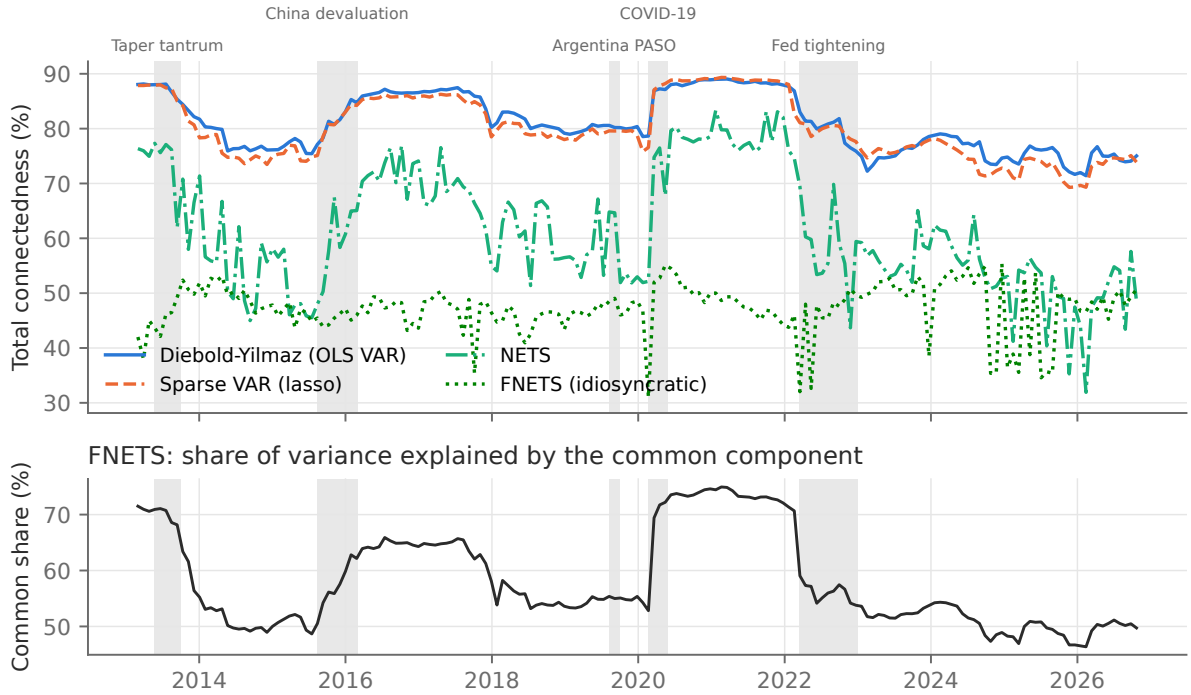
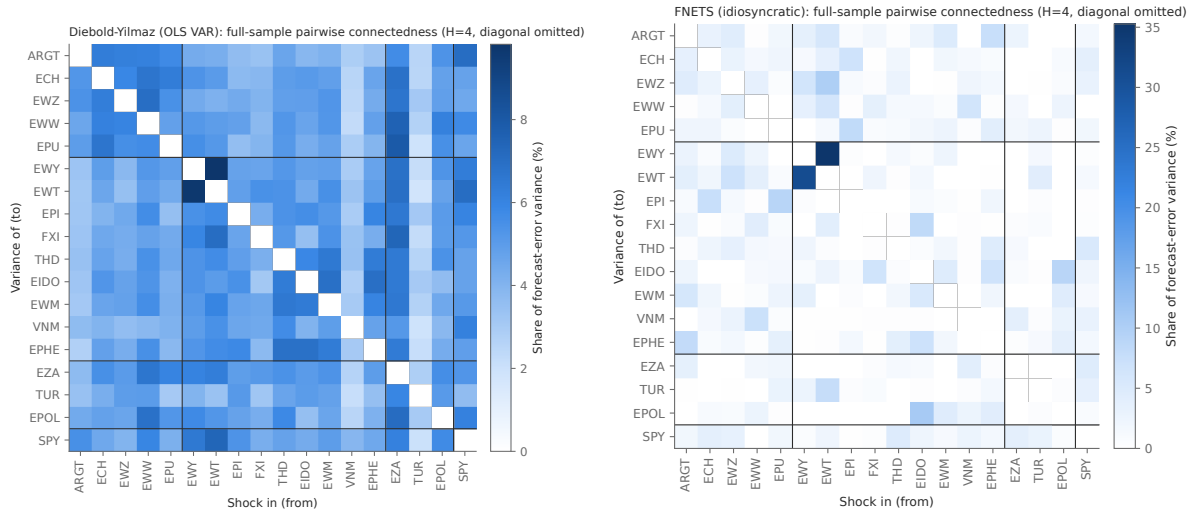


Fig. 2. Top: total connectedness of all methods on a common scale. Bottom: FNETS share of variance explained by the common component.



(a) DY (b) FNETS (idiosyncratic)

Fig. 3. Full-sample pairwise connectedness, countries grouped by region (Latin America, Asia, EMEA, global).

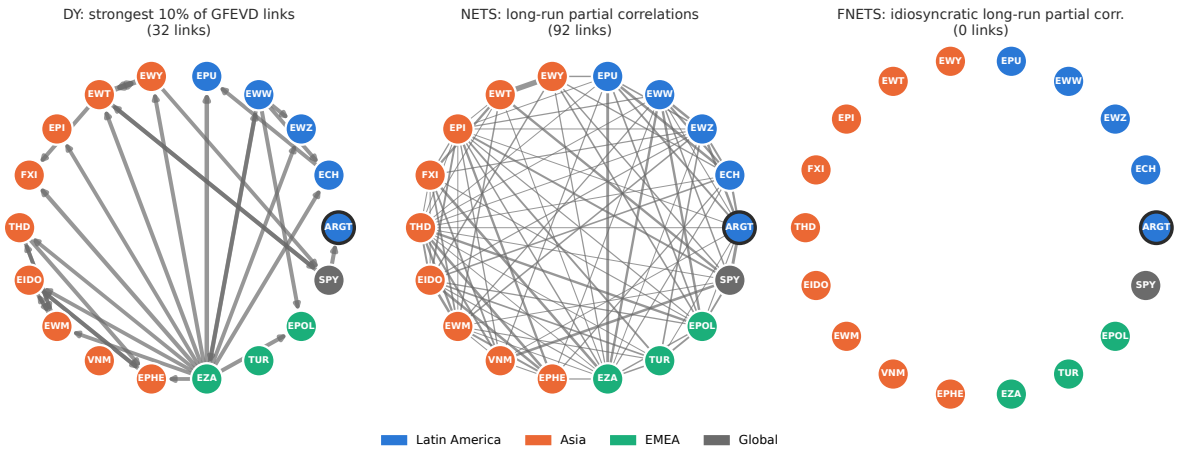


Fig. 4. Full-sample networks. Node colors denote regions; ARGT is outlined. Left: strongest 10% of Diebold–Yilmaz links. Center: NETS long-run partial correlations. Right: FNETS idiosyncratic long-run partial correlations (no links in the full sample).

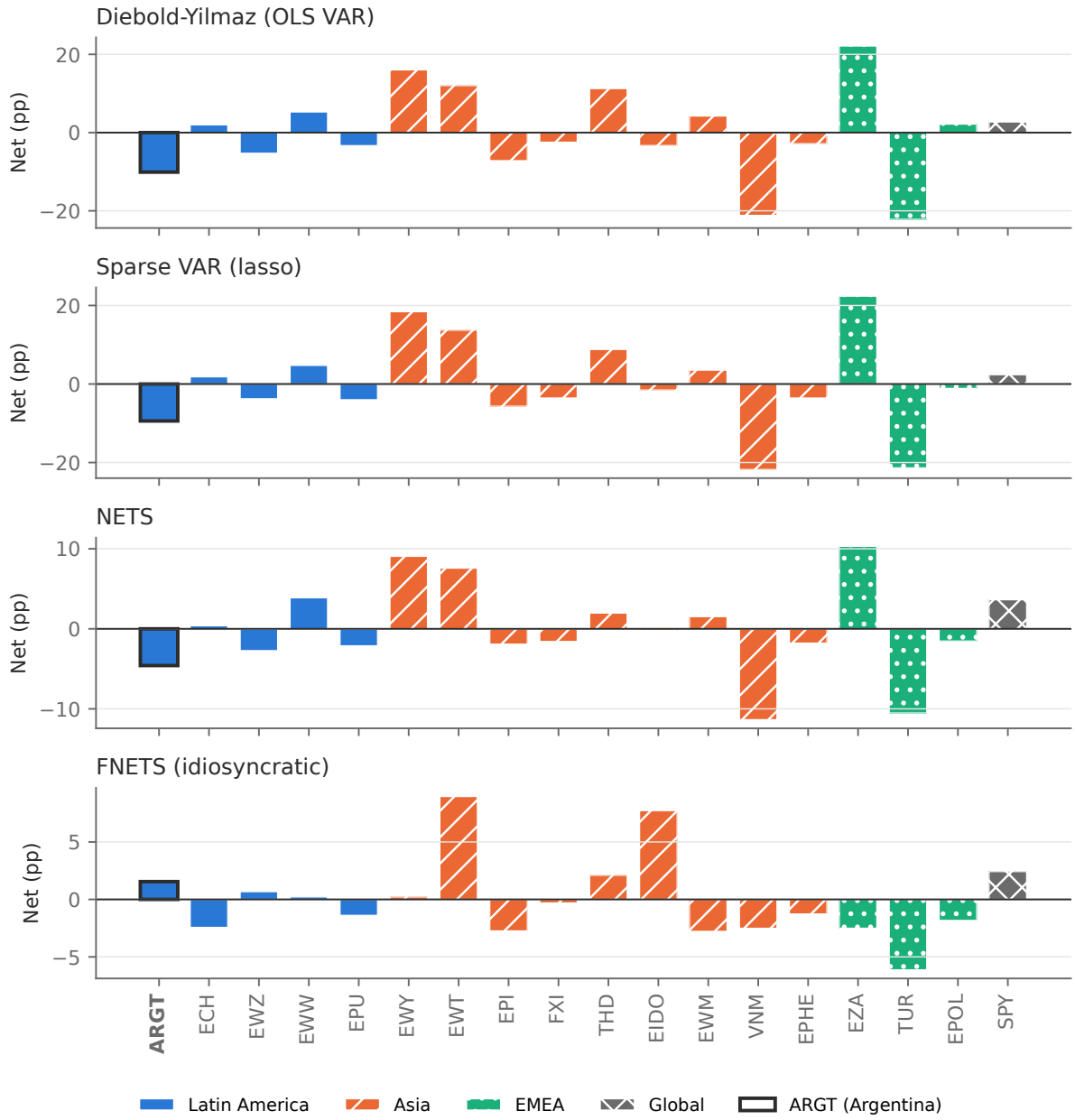


Fig. 5. Average net connectedness by country (transmitters > 0, receivers < 0), in percentage points. ARGT outlined.

6 Discussion

How much of measured connectedness is a common factor? A single factor explains on average 58% of the variance of weekly returns, and removing it lowers total connectedness by about 41% (38–43% across all robustness checks). This number, however, understates the role of the factor. Diebold–Yilmaz connectedness is a share of the forecast-error variance of the returns, whereas FNETS connectedness is a share of the forecast-error variance of the idiosyncratic component only, which is itself about 42% of the total. As a rough calculation, idiosyncratic spillovers account for about $0.47 \times 0.42 \approx 20\%$ of the variance of a typical market, against the 81% that Diebold–Yilmaz attributes to shocks in other markets. Most of what the dense measure reports as connectedness is therefore exposure to the same shock rather than transmission between markets. This is consistent with a global financial cycle (Rey, 2013): the factor is not simply the US market, since excluding the S&P 500 leaves both the common share and the loss of connectedness unchanged, so it is a factor common to emerging markets as a group. It is also consistent with Forbes and Rigobon (2002). The peak of Diebold–Yilmaz connectedness in the windows containing March 2020 coincides with a peak of the common share, while idiosyncratic connectedness shows no peak at all: what looks like contagion in the dense measure is an increase in the weight of the common shock, together with higher volatility, and not an increase in bilateral links. The same reading applies to the decline after 2020, which comes with a smaller common share (58% to 47%) and a slightly higher idiosyncratic connectedness. The high correlation (0.95) between the common share and Diebold–Yilmaz connectedness is partly mechanical, because both increase whenever all pairwise correlations increase; it shows that the two quantities summarize the same feature of the data, not that one causes the other.

Does the method change the picture? Regularization alone does not. The lasso sets almost all lag coefficients to zero and still reproduces Diebold–Yilmaz connectedness (correlation 0.98), and the horizon has virtually no effect. Both facts have the same explanation: at the weekly frequency the moving-average coefficients Ψ_h for $h \geq 1$ are close to zero, so the generalized variance shares in equation (1) reduce to approximately ρ_{ij}^2, the squared correlations, normalized by row. In this setting Diebold–Yilmaz connectedness is close to a normalized squared-correlation matrix, the kind of pairwise and contemporaneous measure that Diebold and Yılmaz (2014) set out to improve on. The approximating model adds little dynamic content here because the data contain little. NETS keeps the same time pattern at a lower level because it regularizes the innovation concentration matrix, not only the lags, and sets weak partial correlations to zero; the connectedness it removes is mostly diffuse, small links. FNETS measures a different object, and its low agreement with the other methods is expected by construction. The choice of method should therefore follow the purpose. For monitoring how tightly markets move together, Diebold–Yilmaz or the lasso VAR are preferable: they are smooth, robust to every specification we tried, and simple to compute. For portfolio diversification across emerging markets, the relevant quantity is the common-factor share, because that part of the co-movement cannot be diversified away within this universe. For the study of contagion, only a factor-adjusted measure such as FNETS addresses the right question, but in our setting it is also the least reliable estimate.

Transmitters and receivers. South Africa, Korea and Taiwan are net transmitters and Turkey, Vietnam and Argentina net receivers in the three systems estimated on returns, with a near-identical ranking (Spearman 0.96–0.99). These directions should not be read as evidence of lead–lag transmission. The Granger networks keep at most 2.5% of the possible lag links, so the data contain very little directional information; with negligible dynamics, the variance shares are close to symmetric squared correlations, and net connectedness mainly reflects the row normalization, that is, which markets are more or less correlated with the rest of the system.